

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		award (2) for 0.6 accept any value in the range 0.57-0.63 if answer incorrect award (1) for correct Δy or Δx value taken from tangent		2		2	2	
		(ii)		award (1) for either of following rate is (directly) proportional to concentration rate halves as concentration halves accept converse neutral answer – rate falls as concentration falls			1	1	1	
	(b)			measure change in colour with time / use a colorimeter (1) colour becomes darker as bromine is formed (1) OR measure change in pH with time / use a pH probe (1) pH rises as H^+ used up (1) reference to time may be linked to the change or the explanation but must be included for both marks to be awarded	1		1	2		2
	I			0.4 mol dm^{-3}			1	1	1	
				Question 9 total	1	2	3	6	4	2